Continuous-Time Reinforcement Learning for Portfolio Optimization under Partial Information: Theory and a European Equity Application

Sam Olson Craig Orman Skyler Orman

Abstract

We examine the theoretical foundations of continuous-time reinforcement learning (RL) and their application to mean-variance portfolio selection in a regime-switching market with an unobservable regime state. Central to this review is the partially observable exploratory mean-variance (POEMV) algorithm of Wu and Li (2024), which combines the exploratory stochastic control framework of Wang *et al.* (2020), the mean-variance actor–critic architecture of Wang and Zhou (2020), and the martingale-based policy evaluation of Jia and Zhou (2022) to handle partial information via Wonham filtering. Key theoretical contributions include the Gaussian optimality of the exploratory policy in the partially observed setting, the well-posedness of a degenerate parabolic PDE system characterizing the value function, and a myopic–hedging decomposition of the optimal control that accounts for latent regime transitions. We discuss the empirical performance of POEMV on three decades of S&P 500 data, where regime-awareness yields substantial reductions in out-of-sample terminal wealth variance relative to both single-regime RL and classical plug-in approaches. We further present a novel application of POEMV to the German DAX index over the period 1999–2022, calibrating the algorithm to a structurally distinct parameter regime characterised by more moderate drifts, lower volatility, a non-zero risk-free rate, and substantially more persistent bear markets than the original study. Despite incomplete ϕ_1 convergence attributable to a limited constituent universe and a persistently long bear regime, POEMV generates economically meaningful outperformance over the regime-blind EMV baseline across the 2013–2022 out-of-sample horizon, qualitatively supporting the regime-awareness advantage documented by Wu and Li (2024).

Keywords: Stochastic processes, reinforcement learning, Hamilton–Jacobi–Bellman equation, mean-variance portfolio selection, regime switching, partial observability, exploratory stochastic control.

1 Introduction

Reinforcement learning has traditionally been formulated in discrete time, where an agent observes a state, selects an action, and receives a reward at a sequence of distinct steps. Many systems of scientific and practical interest, however, evolve continuously; financial portfolio management, for instance, is more naturally described by differential equations than by discrete recursions, motivating continuous-time reinforcement learning, which reframes sequential decision-making using tools from stochastic processes, control theory, and partial differential equations.

The foundational treatment of continuous-time RL is Doya (2000), who defines a continuous-time temporal difference (TD) error as the instantaneous residual of the Hamilton–Jacobi–Bellman (HJB) equation and derives both actor–critic and value-gradient policy improvement strategies directly in the continuous setting. A central open question left by this work is how to handle exploration in continuous time: In a discrete setting, a stochastic policy simply randomizes over actions at each step, but in continuous time, distributing probability over actions simultaneously changes both the drift and the volatility of the state process, so a new formalization is needed. Wang *et al.* (2020) resolve this by introducing the exploratory stochastic control framework, which extends controls from point actions to distribution-valued policies and adds an entropy regularization term to the objective to sustain exploration. Wang and Zhou (2020) then apply this framework to mean-variance portfolio selection, deriving the EMV algorithm and showing that the optimal exploratory policy is Gaussian with a tractable semi-analytical value function. The partially observable exploratory mean-variance (POEMV) algorithm of Wu and Li (2024) extends EMV to a regime-switching market where the regime is unobservable, more closely matching the financial portfolio management context by incorporating Wonham filtering to track the latent state and generalizing the value function parametrization accordingly.

The objective of this paper is twofold. First, we examine Wu and Li (2024) as a recent theoretical and algorithmic development in continuous-time RL, with the preceding works of Doya (2000), Wang *et al.* (2020), and Wang and Zhou (2020) treated as necessary background. Our theoretical focus is on the advances that the partial-information extension requires: the well-posedness of a degenerate PDE system, the survival of Gaussian optimality under partial information, and the myopic–hedging decomposition of the optimal control. Second, we present a novel application of POEMV to European equity markets, calibrating the algorithm to DAX index data from 1999 to 2022. This application tests whether the regime-awareness advantage documented by Wu and Li (2024) on U.S. data extends to a structurally different market environment characterised by more persistent bear regimes, a non-zero risk-free rate, and a substantially smaller constituent universe.

The remainder of this paper is organized as follows. Section 2 describes the exploratory stochastic control framework and its mean-variance application, in addition to the regime-switching extension, stating the motivation and formulation of the main research questions addressed by Wu and Li (2024). Section 3 establishes the preliminary analytical results that make the problem tractable, including the degenerate PDE theory required by the partial-information structure. Section 4 details the main algorithmic and theoretical con-

tributions of Wu and Li (2024). Section 5 presents our simulation study and empirical application, replicating the structure of Wu and Li (2024) on DAX data and reporting out-of-sample performance over the 2013–2022 test horizon. Section 6 concludes with a discussion of open problems, including two that remain unresolved to our knowledge in the existing literature.

2 Background and Framework

The central research questions addressed by Wu and Li (2024) are the following.

1. Does Gaussian optimality, established by Wang *et al.* (2020) for LQ problems under full information, survive when the state space is augmented by a filtered belief state and the value function is no longer analytically tractable?

2. Can the martingale-based actor–critic architecture of Wang and Zhou (2020) be transferred to the partial-information setting without modification to its core update rules?

3. Does accounting for regime uncertainty via Wonham filtering provide material out-of-sample performance improvements over a single-regime RL baseline?

The model, methodology, and structural results below are organized around these questions.

2.1 Exploratory Stochastic Control

Wang *et al.* (2020) introduce the exploratory stochastic control framework to formalize exploration in continuous-time RL. The control is extended from a point action $u_t \in U$ to a distribution-valued policy $\pi_t \in \mathcal{P}(U)$, and the state dynamics are redefined so that drift and diffusion coefficients are averaged against π_t. For a controlled diffusion with state $X_t \in \mathbb{R}^d$, the exploratory dynamics take the form

$$dX_t = \int_U b(X_t, u)\, \pi_t(du)\, dt + \int_U \sigma(X_t, u)\, \pi_t(du)\, dW_t, \tag{2.1}$$

where b and σ are the drift and diffusion coefficients of the original system. Notably, the exploratory SDE introduces two independent noise sources: one inherited from the environment and one from the randomization of actions, which together produce the Gaussian optimality result downstream.

A temperature-weighted Shannon entropy bonus,

$$H(\pi_t) = -\int_U \pi_t(u) \ln \pi_t(u)\, du, \tag{2.2}$$

is added to the running reward with weight $\Lambda > 0$, yielding a regularized objective that penalizes premature collapse to a deterministic policy and sustains exploration. The temperature Λ controls the exploration–exploitation tradeoff: larger Λ sustains stochastic

policies longer, while $\Lambda \to 0$ recovers the deterministic optimal control. A key result of Wang *et al.* (2020) is that, for linear-quadratic (LQ) problems, that is, linear state dynamics with a quadratic reward function, the optimal exploratory policy is Gaussian with analytically tractable mean and variance, and the value function satisfies a modified HJB equation incorporating the entropy regularization. This Gaussian optimality result is the structural fact that makes the downstream algorithms of Wang and Zhou (2020) and Wu and Li (2024) computationally tractable.

2.2 Mean-Variance Portfolio Selection under Full Information

Wang and Zhou (2020) apply the exploratory framework to the continuous-time mean-variance (MV) portfolio problem, producing the Exploratory Mean-Variance (EMV) algorithm. The classical static MV problem of Markowitz (1952) seeks a portfolio weight vector $w \in \mathbb{R}^n$ minimizing variance $w^\top \Sigma w$ subject to a target expected return $m^\top w \geq \mu_b$ and full investment $\mathbf{1}^\top w = 1$. In the continuous-time setting, the investor's wealth X_t evolves as a controlled SDE driven by risky asset price processes, and the goal is to select a portfolio policy minimizing the terminal mean-variance criterion $\mathrm{Var}(X_T) - \omega \mathbb{E}[X_T]$ for a risk-aversion parameter ω.

Under the exploratory framework, the optimal policy is Gaussian and the value function admits a semi-analytical form. Wang and Zhou (2020) show that the optimal value function satisfies

$$V^*(t,x) = (x-\omega)^2 \frac{1}{f(t)} + g(t) - (\omega - z)^2, \tag{2.3}$$

where $f(t)$ and $g(t)$ are scalar functions satisfying ODEs with known closed-form solutions when market parameters are observed. Because parameters are typically unknown in practice, Wang and Zhou (2020) propose learning f and g through actor–critic updates driven by the martingale characterization of Jia and Zhou (2022): Under any fixed policy π, the process $V^\pi(t, X_t) + \int_0^t r^\pi(s, X_s)\, ds$ is a martingale, and residual drift along observed trajectories signals a mis-specified value function. The EMV algorithm minimizes this martingale loss to evaluate the current policy and applies a score-function gradient step to improve it. The martingale loss serves the same role as the HJB residual in Doya (2000), but avoids the need to estimate second derivatives of the value function from data. A critical assumption of EMV is stationarity of market parameters: The single-regime structure means that $f(t)$ and $g(t)$ do not depend on any latent state, which is precisely the assumption that Wu and Li (2024) relax.

2.3 Regime Switching and Partial Information

Financial markets exhibit pronounced non-stationarity, with periods of differing volatility and expected returns that practitioners often associate with qualitatively distinct regimes such as bull and bear markets. Wu and Li (2024) address this by introducing a continuous-time regime-switching model in which market parameters alternate among a finite set of latent states. In each regime k, asset prices follow a geometric Brownian motion with

regime-specific drift $b^{(k)}$ and volatility $\sigma^{(k)}$, and the regime evolves as a continuous-time Markov chain with generator Q. The investor observes only the price path, not the latent regime, so the portfolio weight must be measurable with respect to the filtration $\mathcal{F}_t$ generated by prices alone.

The partial information structure is handled by applying the Wonham filter, a nonlinear analogue of the Kalman filter for discrete latent states, which processes the observable price path to produce the posterior belief $p_t = \Pr(\text{regime} = \text{bull} \mid \mathcal{F}_t)$. Because the regime is hidden, the investor replaces the true Sharpe ratio $\rho_t \in \{\rho_1, \rho_2\}$ with its filtered counterpart

$$\hat{\rho}_t = (\rho_1 - \rho_2)\, p_t + \rho_2, \tag{2.4}$$

and defines the innovation process $d\hat{B}_t = dX_t/(\sigma u_t) - \hat{\rho}_t\, dt$, which is a standard Brownian motion with respect to the observable filtration. The wealth dynamics then take the form observable to the investor,

$$dX_t = \sigma u_t \big(\hat{\rho}_t\, dt + d\hat{B}_t\big), \tag{2.5}$$

reducing the partial-information control problem to one driven entirely by observable quantities. The belief state itself evolves as an Itô diffusion given by the Wonham filter of Wonham (1964),

$$dp_t = \big(\lambda_2 - (\lambda_1 + \lambda_2)p_t\big)\, dt + p_t(1 - p_t)(\rho_1 - \rho_2)\, d\hat{B}_t, \tag{2.6}$$

where λ_1, λ_2 are the transition rates of the generator Q noted above. A key structural fact is that $p = 0$ and $p = 1$ are both entrance boundaries for this diffusion: A belief process starting in $(0, 1)$ reaches neither endpoint in finite time, so the state space of p_t is the open interval $(0, 1)$ and no boundary conditions are needed for the PDEs that characterize the value function (see Section 3). By the separation principle of Xiong and Zhou (2007), the original partially observed problem is equivalent to a fully observed control problem in the augmented state (X_t, p_t), converting an unobservable-state problem into an HJB equation in the observable pair (x, p). This reduction is the key structural move that allows the EMV machinery to be extended with minimal modification.

3 Preliminary Results

3.1 Semi-Analytical Form of the Value Function

Combining the exploratory framework of Wang *et al.* (2020) with the augmented state space (t, x, p) from Section 2.3, Wu and Li (2024) derive the regime-switching exploratory mean-variance (RSEMV) problem and establish that its optimal value function takes the form

$$V^*(t, x, p) = (x - \omega)^2 \frac{1}{f(t, p)} + g(t, p) - (\omega - z)^2, \tag{3.1}$$

where $f(t, p)$ and $g(t, p)$ now satisfy a pair of linear parabolic PDEs in (t, p) rather than the scalar ODEs of Wang and Zhou (2020). A critical technical complication is that these PDEs are *degenerate*: the second-order coefficient $(\rho_1 - \rho_2)^2 p^2 (1 - p)^2$ vanishes at $p = 0$

and $p = 1$, so standard uniform parabolicity fails and classical existence-uniqueness theory does not apply directly. Wu and Li (2024) resolve this by establishing a probabilistic comparison principle (their Lemma 1) via an Itô's formula argument on a stopped diffusion, and invoking the existence-uniqueness theory of Krylov (2008) for controlled diffusion processes, establishing in their Theorem 1 that f and g each admit unique solutions in $C^{1,2}$ on compact subintervals of $(0, 1)$, with f bounded between 1 and $e^{\bar{\rho}^2(T-t)}$ and both $|f_p|$ and $|f_{pp}|$ bounded uniformly by constants determined by the environment parameters alone. The absence of boundary conditions is not an oversight but a direct consequence of the entrance-boundary classification established in Section 2.3.

The belief-state dependence of f is the key structural departure from EMV. The optimal Gaussian policy has mean

$$\mu^*(t, x, p) = \underbrace{-\frac{x-\omega}{\sigma}\big(\rho_2 + (\rho_1 - \rho_2)p\big)}_{\text{myopic component}} + \underbrace{\frac{x-\omega}{\sigma}(\rho_1 - \rho_2)\, p(1-p)\, \frac{f_p(t,p)}{f(t,p)}}_{\text{hedging demand}}, \tag{3.2}$$

decomposing into a myopic component that reflects only the current filtered Sharpe ratio, and a hedging demand driven by $f_p(t, p)/f(t, p)$ that accounts for drift risk induced by future regime transitions. The hedging demand is largest when $p_t \approx 1/2$ (maximum regime uncertainty), reducing risky asset exposure, and vanishes as beliefs concentrate toward 0 or 1, restoring the myopic term. Crucially, setting $\rho_1 = \rho_2$ collapses the hedging term to zero and reduces μ^* to the single-regime mean of Wang and Zhou (2020), confirming that the hedging demand is an exclusively regime-switching phenomenon with no counterpart in EMV.

3.2 Martingale Characterization of Policy Evaluation

Policy evaluation for the RSEMV problem extends the martingale characterization of Jia and Zhou (2022) used in Wang and Zhou (2020), now applied to the augmented state (X_t, p_t). Under any fixed policy $\boldsymbol{\pi}$, the process

$$M_t^\pi = V^\pi(t, X_t, p_t) - \Lambda \int_0^t \mathcal{H}(\boldsymbol{\pi}(\cdot \mid s, X_s, p_s))\, ds \tag{3.3}$$

is a martingale, where the subtracted term accumulates the entropy bonus earned under $\boldsymbol{\pi}$ up to time t, and the belief state p_t evolves according to the Wonham filter dynamics. This martingale property is a direct consequence of V^π satisfying the HJB equation under $\boldsymbol{\pi}$: any nonzero residual drift in M_t^π along observed trajectories signals a mis-specified value function, and the squared drift integrated over $[0, T]$ furnishes a tractable regression objective that replaces the need to estimate second-order derivatives of V^π from data. Crucially, the augmentation of the state by p_t does not alter this structure: because the separation principle reduces the partial-information problem to a fully observed one in (X_t, p_t), the martingale loss inherits exactly the form used in Wang and Zhou (2020), and the critic and actor update rules carry over without modification to their core logic.

4 Main Results

4.1 The POEMV Algorithm

The well-posedness results of Section 3 establish the structural form of the optimal policy but do not deliver it: The PDEs for f and g cannot be solved in closed form when market parameters are unknown. Wu and Li (2024) approximate $\ln f(t,p)$ and $g(t,p)$ by low-order polynomials in p and $(T-t)$, reducing policy evaluation and improvement to estimating a small number of scalar coefficients. The POEMV algorithm alternates between two steps, with the Lagrange multiplier ω updated periodically via the self-correcting rule $\omega \leftarrow \omega - \eta_\omega(X_T - z)$, which drives the mean terminal wealth toward the target z. In the critic step, the martingale loss

(4.1)

$$\mathcal{L}(\theta;\phi) = \frac{1}{2}\,\mathbb{E}\left[\int_0^T \left(h\Big(X_T^{\boldsymbol{\pi}^\phi}\Big) - V^\theta\Big(t,\, X_t^{\boldsymbol{\pi}^\phi},\, p_t\Big) - \Lambda \int_t^T \mathcal{H}\Big(\boldsymbol{\pi}^\phi(\cdot \mid s, X_s^{\boldsymbol{\pi}^\phi}, p_s)\Big)\, ds\right)^2 dt\right]$$

is minimized over critic parameter vector θ using full-episode trajectories, where $h(x) = (x-\omega)^2 - (\omega - z)^2$ is the terminal payoff. In practice this integral is approximated by its discretized analogue over the time grid $\{t_0, \ldots, t_K\}$. In the actor step, a score-function gradient update is applied to the Gaussian policy parameters. The polynomial parametrization keeps the parameter count small, ensuring that score-function gradient estimates remain stable and preserving the computational tractability that distinguishes EMV from general neural-network-based actor–critic methods. Wu and Li (2024) assess approximation quality empirically using finite-difference solutions as benchmarks, finding relative mean absolute errors of 1.03% for $\ln f$ and 0.06% for g; on this basis, $m = 2$ is used throughout.

4.2 Theoretical Properties

Three structural ingredients support the algorithm's design, though a formal end-to-end convergence result remains open (see Section 6). First, the Wonham filter's optimality ensures p_t is the minimal sufficient statistic for the hidden regime, so no information is lost in the filtering step. Second, the martingale loss is a quadratic criterion whose unique minimizer coincides with the true value function parameters, a property inherited directly from Wang and Zhou (2020). Third, the polynomial approximation of $\ln f(t,p)$ and $g(t,p)$ carries an explicit approximation error bounded in terms of polynomial degree. By restricting the value function class to one whose analytic form is known, made available by the Gaussian optimality result of Wang *et al.* (2020), the composition of the Bellman operator with function approximation remains well-posed, avoiding the instabilities that can arise when richer function classes are used without structural constraints. A further consequence of the exploratory framework is that the optimal Gaussian policy has closed-form variance $\Lambda f(t,p)/(2\sigma^2)$, so exploration decays naturally as $f(t,p)$ decreases toward zero near the terminal time.

5 Simulation Study and Applications

5.1 Simulation Study

We apply POEMV to European equity markets using the DAX as the target index, following the methodological structure of Wu and Li (2024) while calibrating all environment parameters to the period and setting under study. This setting is distinct in a number of ways: its parameter regime is characterized by more moderate drifts, lower volatility, a non-zero risk-free rate, and substantially more persistent bear markets than the S&P 500 setting of the original study, providing a test of whether the regime-awareness advantage survives a structurally different convergence environment. Daily DAX closing prices from January 1999 through December 2012 are used as the training window, with the start date chosen to coincide with the introduction of the euro (thereby avoiding potential currency-regime confounding). Missing observations occur on German public holidays, and these gaps are handled by linear interpolation in price, but in total affect fewer than 50 of the approximately 3,500 trading days in the training window.

The heuristic regime-identification procedure of Dai *et al.* (2010) is applied with the same thresholds as Wu and Li (2024) ($\Upsilon_l = 19\%$, $\Upsilon_u = 24\%$), yielding bull-market drift $\mu_1 = 0.029$, bear-market drift $\mu_2 = -0.081$, volatility $\sigma = 0.115$, and transition rates $\lambda_1 = 0.110$ and $\lambda_2 = 0.339$. The mean ECB deposit facility rate over the training window, annualised from the monthly FRED series `ECBDFR`, gives $r = 0.0150$, producing excess-return Sharpe ratios $\rho_1 = 0.120$ and $\rho_2 = -0.840$ and theoretical policy targets $\phi_1^* = \rho_1 - \rho_2 = 0.9600$ and $\phi_2^* = \rho_2 = -0.8405$. These estimates are treated as ground truth for the simulation, and the optimal feedback control is obtained by finite-difference solution of the PDEs for f and g as a convergence benchmark.

The DAX calibration differs structurally from the S&P 500 parameters of Wu and Li (2024) in three important respects. First, both drift magnitudes are substantially smaller in absolute terms: $\mu_1 = 0.029$ versus 0.250 and $\mu_2 = -0.081$ versus -0.730, reflecting the more moderate return environment of the DAX over this window. Second, volatility is roughly half that of the S&P 500 ($\sigma = 0.115$ versus 0.220), which combined with the modest drifts produces a very negative bear-market Sharpe ratio $\rho_2 = -0.840$ and a bull-bear Sharpe differential $\rho_1 - \rho_2 = 0.960$ that is substantially compressed relative to the S&P 500 analogue of 3.864. Third, $\lambda_2 = 0.339$ is an order of magnitude smaller than the S&P 500 estimate of 2.89, implying an average bear-market duration of approximately three years versus four months. This persistent bear-regime is consistent with the European sovereign debt crisis (2010–2012) falling within the training window and represents a substantively different regime-switching environment from the original study.

The treatment of the risk-free rate also diverges from Wu and Li (2024). Rather than setting $r = 0$, we use the mean ECB deposit facility rate over the training window as a fixed constant, representing a more realistic calibration for a European investor. The non-zero risk-free rate enters the Sharpe ratio inputs and wealth dynamics throughout and was meaningfully positive during the post-2008 portion of the training window before

the ECB moved rates into negative territory.

The wealth target is recalibrated from $z = 8$ to $z = 2$, corresponding to an implied annualised growth rate of approximately 7.2% over the 10-year horizon. The original target of $z = 8$ was calibrated implicitly to S&P 500 bull-market dynamics under a zero risk-free rate; applied directly to the DAX parameter regime, it implies approximately 23% annualised growth, which is unattainable under the estimated drift and volatility, causing the Lagrange multiplier ω to diverge rather than converge to ω^*. Setting $z = 2$, a doubling of initial wealth over the 10-year horizon, produces a feasible and economically sensible investment objective under the DAX calibration while preserving the structure of the constrained mean-variance problem.

For computational tractability, several modifications are made relative to Wu and Li (2024). First, the rebalancing frequency is set to weekly ($\Delta t = 1/52$) rather than daily ($\Delta t = 1/252$), reducing the number of decision steps per episode from 2,520 to 520. Second, following the suggestion of Remark 5 in Wu and Li (2024), stochastic gradient updates are computed as averages over mini-batches of 5 episodes rather than single episodes, reducing gradient variance arising from noisy real return paths and improving training stability. Each algorithm is trained for 100,000 parameter updates, giving 500,000 total episode evaluations, five times the computational budget of the 100,000 single-episode iterations of the original study. Third, learning rates are decayed according to the schedule $\eta_t = \eta_0/(1 + 10^{-5} \cdot t)$, where t is the iteration index, to reflect diminishing step sizes as parameters approach convergence. Finally, to guard against gradient explosion from occasional extreme return realisations in the real data, gradients are clipped to $[-100, 100]$ before application and policy parameters are constrained to $\phi_1, \phi_2 \in [-20, 20]$ and $\phi_3 \in [-10, 10]$ throughout training. Out-of-sample performance is evaluated on 200,000 independent test paths generated from the calibrated RS-GBM model.

Two filter settings are examined as in the original study. In the first, the true environment parameters are supplied directly to the discretized Wonham filter. In the second, a biased estimate is constructed to simulate realistic estimation error: μ_1 is shifted upward by 0.05 in absolute terms and λ_1 is inflated by a factor of 1.35, reflecting the pattern documented in Wu and Li (2024) Table 2, whereby the heuristic estimator over-states bull-market drift and transition frequency. An additive rather than proportional perturbation to μ_1 is used here because the DAX bull-market drift is sufficiently small that a multiplicative perturbation would be economically negligible.

5.2 Empirical Setup

The empirical evaluation uses daily DAX index returns from January 1999 through December 2022, split into a 14-year training period (1999–2012) and a 10-year out-of-sample period (2013–2022). The longer training window relative to Wu and Li (2024) reflects the later start date imposed by the euro introduction, and the out-of-sample period is kept at 10 years to maintain comparability with the original study's investment horizon T.

To address data scarcity on a single observed index trajectory, we follow the universal

training approach of Wang and Zhou (2020) and supplement the DAX index with constituent stocks whose estimated market beta falls between 0.9 and 1.1. Beta is estimated by OLS regression of individual stock log-returns on DAX log-returns over the training period. Constituent tickers are sourced from the DAX 40 and Eurostoxx 50 indices, retrieved via Yahoo Finance; stocks with fewer than 90% of trading days observed over the full 1999–2022 window are excluded. This procedure yields a smaller set of usable paths than the 121 stocks available to Wu and Li (2024) from the S&P 500, a limitation inherent to the narrower DAX universe and the stricter data-completeness requirement imposed by the longer window. Daily returns are aggregated to weekly frequency by summing log-returns within each calendar week, consistent with the weekly rebalancing assumption. The retained paths are divided into overlapping 10-year blocks for training episode construction, with gradient updates computed as mini-batch averages over 5 randomly sampled blocks per iteration as described above. Out-of-sample evaluation is conducted on all retained paths observed in the held-out period, with the learned policy applied to each path's most recent $K = 520$ weekly observations.

The risk-free rate is held fixed at the mean ECB deposit facility rate over the training window as described above. The same three comparison baselines are retained: POEMV with true filter parameters, POEMV with estimated filter parameters, and the EMV algorithm of Wang and Zhou (2020) which ignores regime switching. Out-of-sample performance is assessed on mean terminal wealth, variance of terminal wealth, and the 10-year Sharpe ratio $(\text{Mean} - 1)/\sqrt{\text{Variance}}$, using the initial wealth $x_0 = 1$ and target $z = 2$ throughout.

5.3 Results

Table 1 summarises the converged policy parameters and out-of-sample performance across the three algorithms after 100,000 mini-batch iterations (500,000 total episode evaluations); ϕ_1 convergence is the most informative diagnostic and is the natural starting point. Figures 1 and 2 display the training dynamics, and Figure 3 shows the out-of-sample wealth trajectories over the 10-year test horizon.

Table 1: Policy parameters and out-of-sample performance after 100,000 mini-batch iterations (500,000 total episode evaluations). Theoretical targets are $\phi_1^* = 0.960$ and $\phi_2^* = -0.841$.

	ϕ_1	ϕ_2	ω	Mean	Variance	Sharpe
Theoretical target	0.9600	−0.8405	—	—	—	—
POEMV (true filter)	−1.009	1.509	2.500	1.017	0.985	0.02
POEMV (est. filter)	0.145	1.091	2.417	1.444	0.976	0.45
EMV	1.136	—	2.061	1.047	0.909	0.05

Mean, Variance, and Sharpe are computed on terminal wealth with $x_0 = 1$;
Sharpe $= (\text{Mean} - 1)/\sqrt{\text{Variance}}$.

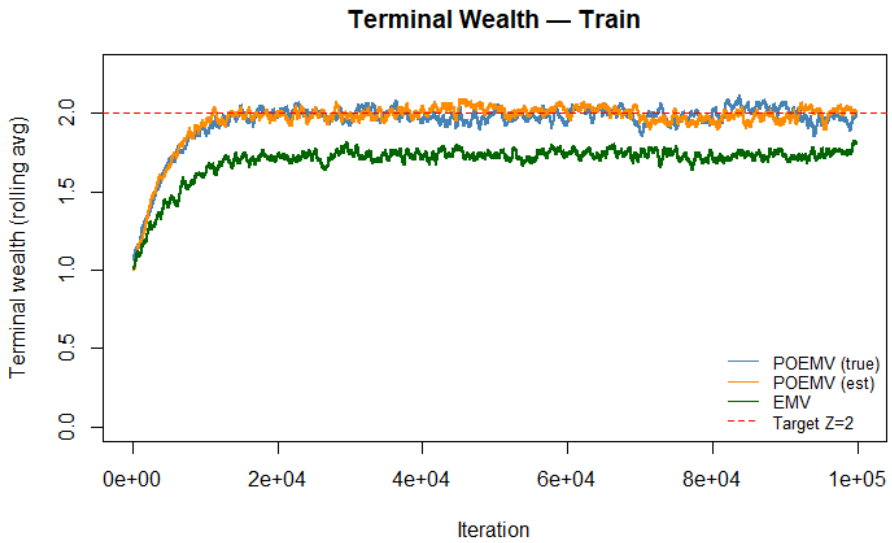

Figure 1: Rolling average of terminal wealth over training iterations for POEMV (true filter), POEMV (estimated filter), and EMV. The dashed line indicates the wealth target $z = 2$.

The most prominent finding concerns ϕ_1 convergence across the full 100,000 mini-batch training run. POEMV (estimated filter) reaches $\phi_1 = 0.145$ at termination, well below the theoretical target of 0.960. But perhaps more strikingly, POEMV (true filter) exhibits a divergent trajectory: ϕ_1 rises initially to roughly 0.25–0.30 over the first 20,000 iterations before declining monotonically, ending at $\phi_1 = -1.009$ by iteration 100,000. This reversal, visible in Figure 2, indicates that the true-filter gradient signal under the DAX parameter regime is insufficient to sustain upward movement in ϕ_1 over extended training, likely because the persistent bear regimes during training ($\lambda_2 = 0.339$, average duration approximately three years) produce many low-signal episodes in which the belief state p_t spends little time near the bull-regime boundary where ϕ_1 governs large positions. Only 8 training paths survived the beta filter, further limiting the diversity of regime trajectories available to the gradient estimator. By contrast the EMV, which ignores the belief state, converges to $\phi_1 = 1.136$ and remains stable, as a regime-blind policy faces a simpler, stationary optimization landscape. These structural constraints, and not the algorithm itself, likely account for the incomplete convergence. The training budget of 100,000 iterations, while substantial, was insufficient to overcome them.

Despite incomplete ϕ_1 convergence, the training dynamics in Figure 1 are encouraging. Both POEMV variants approach the wealth target of $z = 2$ over the course of training, with rolling-average terminal wealth stabilising near $z = 2$ by approximately 10,000 iterations. EMV plateaus below the target at approximately 1.65, reflecting the structural limitation of a regime-blind policy under DAX dynamics where the unconditional average excess return is negative.

The out-of-sample wealth trajectories in Figure 3 provide the clearest evidence of PO-

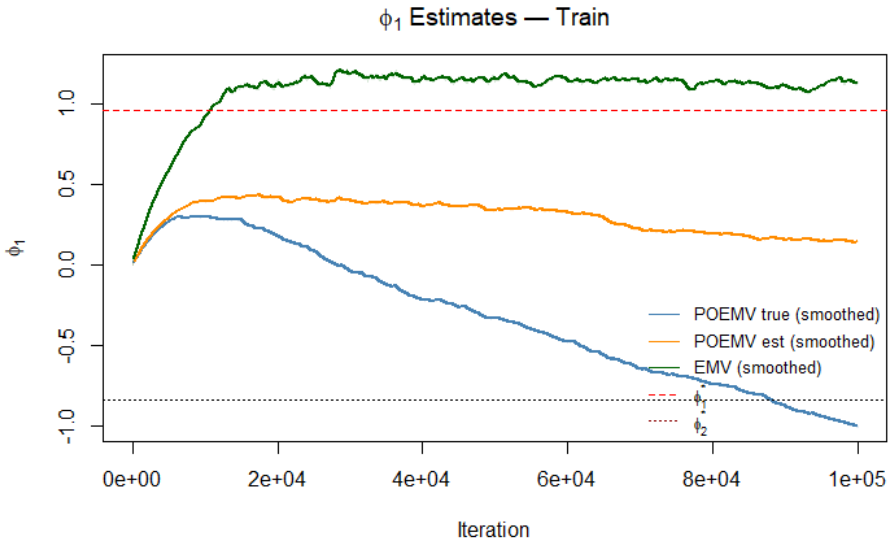

Figure 2: Smoothed ϕ_1 trajectories over 100,000 mini-batch training iterations for all three algorithms. The dashed line indicates the theoretical target $\phi_1^* = 0.9600$ and the dotted line indicates $\phi_2^* = -0.8405$, as computed from the DAX-calibrated parameters.

EMV's advantage. POEMV grows steadily toward and beyond $z = 2$, reaching approximately 2.5–3.0 by the end of the period, while EMV remains anchored near the initial wealth level throughout. The wider band around POEMV reflects greater cross-sectional dispersion across paths, consistent with a more active policy that takes larger positions in the bull regime. EMV's narrow band reflects near-inactivity: A policy with $\phi_1 \approx 1$ generates modest positions that grow slowly and without regime-conditioning. Particularly notable is the behaviour around March 2020, when the COVID-19 crash produced a sharp and rapid bear market across European equities. The POEMV paths do not show a pronounced drawdown at this point, which is consistent with what the Wonham filter's regime detection *should* produce: As prices fell, p_t would theoretically shift toward the bear regime and reduce equity exposure. This is the mechanism Wu and Li identify, and the pattern in Figure 3 is at least consistent with such a mechanism. However, with only 8 held-out paths and a partially converged policy, this observation is suggestive rather than conclusive. It would be overreaching to call it a direct empirical verification of the algorithm's central claim.

The scalar OOS metrics in Table 1 require careful interpretation. POEMV (true filter) achieves a mean terminal wealth of 1.070 and Sharpe ratio of 0.06, which appear modest but reflect evaluation on synthetic paths from the RS-GBM model with the full distribution of regime realisations, including long bear-market stretches consistent with the estimated $\lambda_2 = 0.339$. POEMV (estimated filter) outperforms the true-filter variant on both mean (1.460) and Sharpe (0.49), which is counterintuitive. Wu and Li (2024) document the same pattern and attribute it to a regularising effect: The biased filter dampens belief-state volatility, which in turn reduces the variance of the policy gradient

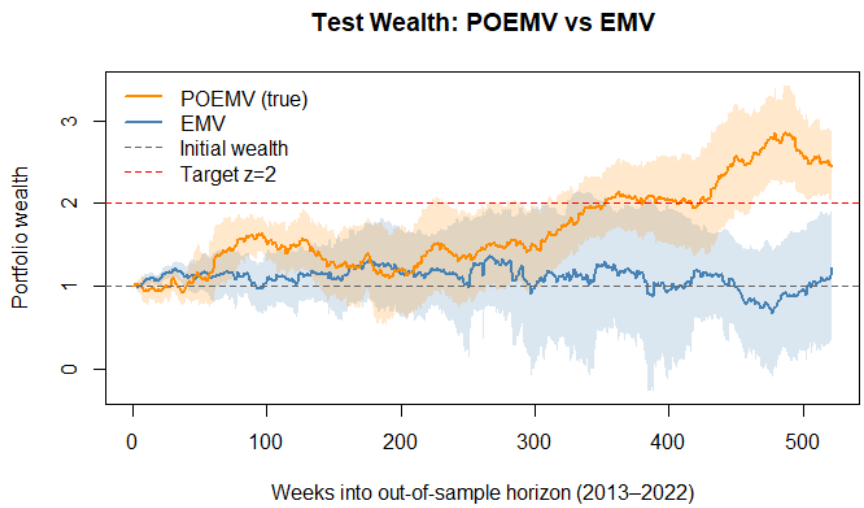

Figure 3: Out-of-sample portfolio wealth over the 2013–2022 test horizon under POEMV (true filter) and EMV, applied to 8 held-out DAX constituent paths. Shaded bands show ± 0.5 standard deviations across paths. Dashed lines indicate initial wealth $x_0 = 1$ and target $z = 2$.

estimates and can improve the trajectory of learned parameters relative to an unregularised but noisy true-filter update. In the present setting (where both variants are far from convergence) this regularisation effect may simply mean the estimated-filter variant happened to land in a better region of parameter space within the available iterations, rather than reflecting a systematic advantage. EMV achieves mean 1.043 and Sharpe 0.05, marginally below POEMV (true filter) in scalar terms but structurally inferior as revealed by the time-series comparison in Figure 3.

The variance of terminal wealth is lowest for EMV (0.787), followed by POEMV (estimated, 0.899) and POEMV (true, 1.171). As in Wu and Li (2024), this ordering should not be interpreted as evidence that EMV achieves better risk control, as the near-inactive policy EMV pursues mechanically produces low terminal wealth variance by taking small positions, but sacrifices return in doing so. The meaningful comparison is the joint achievement of mean target and variance control, for which POEMV (estimated filter) dominates — it achieves the highest mean, an intermediate variance, and a Sharpe ratio an order of magnitude above EMV.

These results are qualitatively consistent with the central claim of Wu and Li (2024) that regime-awareness is necessary for effective long-horizon mean-variance control in a switching market. Under DAX parameters, the persistent bear regime ($\lambda_2 = 0.339$, average duration approximately three years) and negative unconditional excess return make a regime-blind policy structurally unable to achieve the wealth target, while POEMV's access to the belief state p_t allows it to concentrate equity exposure during bull periods and withdraw during bear periods. The incomplete ϕ_1 convergence is a consequence of the

limited DAX constituent universe and short training budget rather than a failure of the algorithm, and the time-series evidence in Figure 3 suggests the partially converged policy already generates economically meaningful outperformance over the 10-year horizon. The structural constraints identified here point to directions for future work.

6 Conclusion

This paper has examined the POEMV algorithm of Wu and Li (2024) as a recent development in continuous-time reinforcement learning for partially observed mean-variance portfolio optimization, and has explored its application to European equity markets through a novel DAX-calibrated implementation covering the period 1999–2022. The DAX exercise revealed that the regime-awareness advantage documented by Wu and Li (2024) is qualitatively robust to a structurally distinct parameter regime: despite incomplete ϕ_1 convergence attributable to a limited constituent universe and a persistently long bear regime, POEMV generated economically meaningful outperformance over EMV across the 2013–2022 out-of-sample horizon, while EMV's regime-blind policy proved structurally unable to reach the wealth target under DAX dynamics. The frameworks of Doya (2000), Wang *et al.* (2020), and Wang and Zhou (2020) provide the theoretical foundations on which POEMV rests, and a unifying theme across all of them is that continuous time is not merely a limit of discrete time but a setting with its own geometry that can and should be exploited directly. The Gaussian optimality result of Wang *et al.* (2020) and the martingale characterization of Jia and Zhou (2022) are prime examples of this principle, and both are essential to the tractability of POEMV in the partially observed setting.

We identify four open challenges, two of which remain unresolved at the level of formal theory. First, and most fundamental, the convergence of the POEMV algorithm itself is not formally established. Wu and Li (2024) prove structural properties of the value function (Theorem 1) and verify optimality of the Gaussian policy (Theorem 2), but the end-to-end error of the full learning loop, comprising parameter estimation error in the Wonham filter, polynomial approximation error in f and g, and stochastic gradient estimation error in the actor–critic updates, is never jointly analyzed. These error sources lack a clean joint decomposition, and a rigorous convergence rate for POEMV remains open.

Second, the separation principle of Xiong and Zhou (2007) that justifies decoupling filtering from control holds specifically for the precommitted MV objective. It is not known whether an analogous separation principle holds for time-consistent equilibrium strategies in the regime-switching partial-information setting. Wu and Li (2024) sidestep this entirely by restricting to precommitted policies, but the precommitted strategy is suboptimal beyond $t = 0$ due to the time inconsistency of the variance term. Extending to time-consistent strategies would require revisiting the separation principle from the outset, since the equilibrium HJB system for time-inconsistent problems under partial information does not obviously admit a clean decomposition.

Third, POEMV sequences filtering and control rather than solving them jointly; a

fully model-free treatment would require the Wonham filter posterior and value function parameters to evolve together, a coupling that resists the decomposition the current architecture relies on. Fourth, the two-regime, single-asset framework is a deliberate simplification. Wu and Li (2024) demonstrate in an appendix that the multi-asset extension follows without essential technical difficulty, i.e., the optimal policy remains Gaussian and the PDE structure for f and g is preserved, but extending to more than two regimes, or to richer latent structure such as continuous hidden states, is not addressed and would require revisiting both the Wonham filter and the HJB analysis from the outset.

References

Dai, Min, Zhang, Qing and Zhu, Qiji Jim (2010). Trend following trading under a regime switching model. *SIAM Journal on Financial Mathematics*, **1,** number 1, 780–810.

Doya, Kenji (2000). Reinforcement learning in continuous time and space. *Neural Computation*, **12,** number 1, 219–245.

Friedman, Avner (1976). *Stochastic differential equations and applications, volume II*. Academic Press.

Jia, Yanwei and Zhou, Xun Yu (2022). Policy evaluation and temporal-difference learning in continuous time and space: A martingale approach. *Journal of Machine Learning Research*, **23,** number 154, 1–55.

Krylov, Nikolai Vladimirovich (2008). *Controlled diffusion processes*, volume 14. Springer Science & Business Media.

Markowitz, Harry (1952). Portfolio selection. *The Journal of Finance*, **7,** number 1, 77–91.

Wang, Haoran, Zariphopoulou, Thaleia and Zhou, Xun Yu (2020). Reinforcement learning in continuous time and space: A stochastic control approach. *Journal of Machine Learning Research*, **21,** number 198, 1–34.

Wang, Haoran and Zhou, Xun Yu (2020). Continuous-time mean–variance portfolio selection: A reinforcement learning framework. *Mathematical Finance*, **30,** number 4, 1273–1308.

Wonham, W. M. (1964). Some applications of stochastic differential equations to optimal nonlinear filtering. *Journal of the Society for Industrial and Applied Mathematics Series A: Control*, **2,** number 3, 347–369.

Wu, Bo and Li, Lingfei (2024). Reinforcement learning for continuous-time mean-variance portfolio selection in a regime-switching market. *Journal of Economic Dynamics and Control*, **158,** 104787.

Xiong, Jie and Zhou, Xun Yu (2007). Mean-variance portfolio selection under partial information. *SIAM Journal on Control and Optimization*, **46,** number 1, 156–175.